AP Precalculus **Unit 4, Lesson 4.2**

Group Activity

Name: ____________________________ Partner: _______________________ Period: _________

Remote-Control Car

A remote-control car moves along a surface. Its position in meters at time t seconds is given by the parametric function

$$f(t) = \big(x(t),\ y(t)\big) = \big(t^2 - 6t + 5,\ -t^2 + 4t\big), \quad 0 \le t \le 6,$$

where $x(t)$ measures meters east of the starting line and $y(t)$ measures meters north.

Part 1 — Building the Table

1. Complete the table. Leave no cell blank.

t (sec)	$x(t) = t^2 - 6t + 5$	$y(t) = -t^2 + 4t$	Position $(x,\ y)$
0	____________	____________	______________
1	____________	____________	______________
2	____________	____________	______________
3	____________	____________	______________
4	____________	____________	______________
5	____________	____________	______________
6	____________	____________	______________

Part 2 — Analyzing the Car's Motion

2. What is the *leftmost* east-position of the car? At what time does this occur? Show your work using the vertex formula.

3. What is the *rightmost* east-position? (Check the endpoints of the domain.)

4. What is the highest north-position and at what time does it occur? Use the vertex formula for $y(t)$. Show your work.

5. When does the car cross the x-axis (where $y = 0$)? Show your algebra, then state the car's coordinates at those moments.

6. When does the car cross the y-axis (where $x = 0$)? Show your algebra, then state the car's coordinates at those moments.

Part 3 — Extension

7. Consider the parametric function $g(t) = (\cos t,\ \sin t)$ for $0 \leq t \leq 2\pi$.

(a) What are the maximum and minimum values of $x(t) = \cos t$? What are the horizontal extrema of this curve?

(b) What are the maximum and minimum values of $y(t) = \sin t$? What are the vertical extrema?

(c) Find all values of t in $[0, 2\pi]$ where $x(t) = 0$. What are the y-intercepts of the curve?